Gradient descent

Study notes | One topic, five subtopics and one figure

1. Purpose

Gradient descent is an optimisation algorithm. It is used, for example, to adjust the parameters of a machine learning model. A loss function measures how poorly a chosen set of parameters performs on the learning task. The objective is to find parameters with a lower loss.

2. Gradient direction

For a differentiable loss function, the gradient points in the direction of steepest local increase in loss, using the usual Euclidean notion of direction. It does not point towards the steepest decrease. In one dimension, the gradient is the derivative of the loss with respect to the parameter.

3. Parameter updates

Gradient descent updates parameters in the direction opposite to the gradient. For one parameter w, the update is:

w_new = w_old - learning_rate * gradient

Here, the gradient is evaluated at the current parameter value and the learning rate is positive. Subtracting a positive gradient decreases w. Subtracting a negative gradient increases w.

Scope

These notes describe ordinary gradient descent on a differentiable loss. They do not cover momentum, adaptive optimisers or stochastic sampling. Mathematical correctness and comparison with these notes should be kept separate from whether an explanation uses identical wording.

Step size and visual example

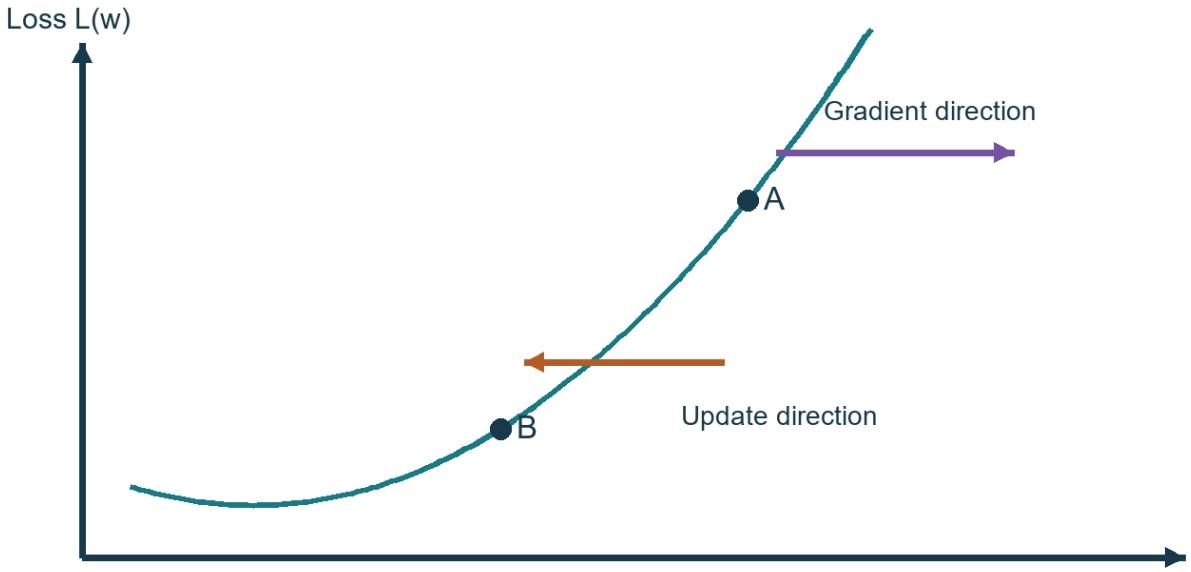
4. Learning rate

The learning rate scales the size of the parameter update. For a fixed non-zero gradient, a larger learning rate produces a larger step. An excessively large learning rate can overshoot a useful region or cause the loss to increase. A very small learning rate can make progress slow.

5. Objective and limitations

The aim of the updates is to reduce the loss over time. This is an objective, not a guarantee that every step reduces loss. Gradient descent does not generally guarantee finding a global minimum for every loss function.

Figure 1. A single update on an illustrative loss curve



One-dimensional illustration; positions are schematic.

Read the labelled positions and arrows in the figure. Explain how the update direction relates to the gradient direction, and compare the losses at the two labelled positions.